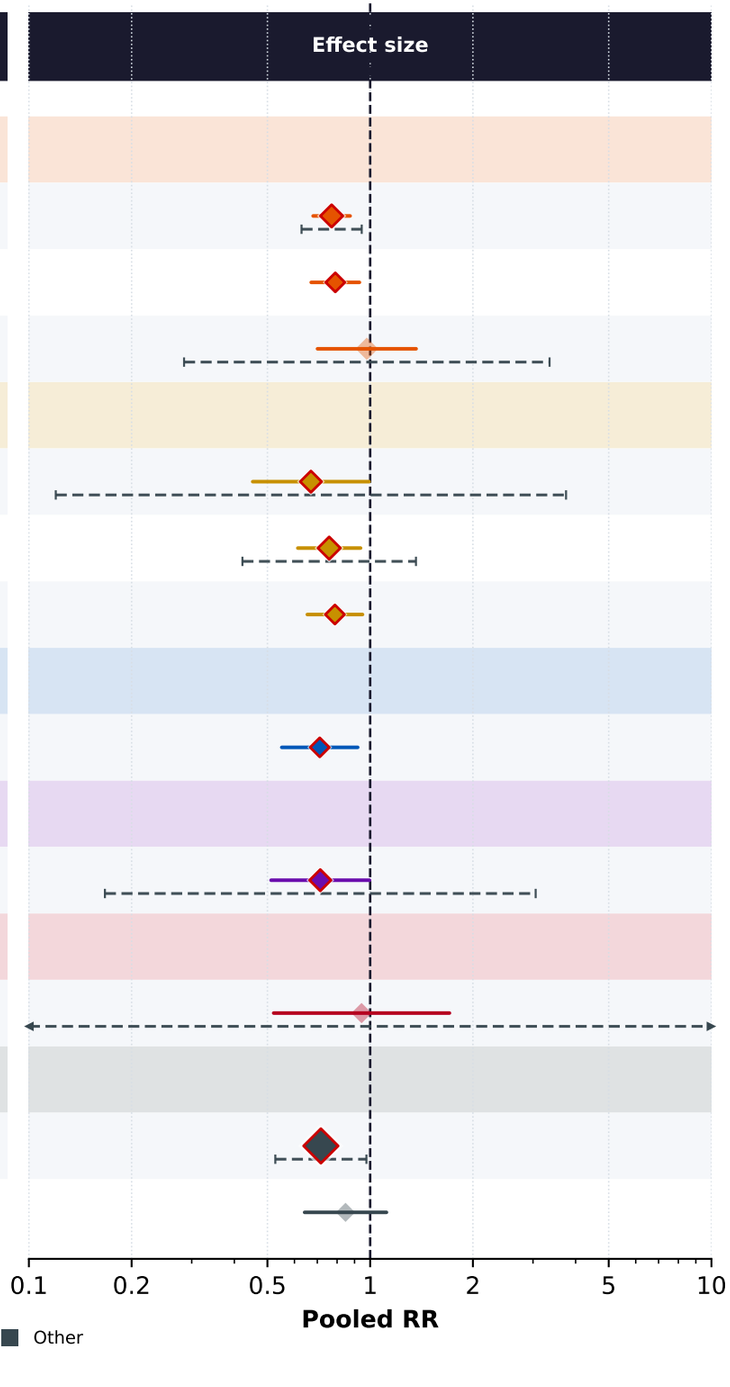
Exposures negatively associated with renal cell carcinoma risk
Meta-analysis of dietary-intake studies

Exposure	Estimates	N	Cases	Pooled RR (95% CI)
CAROTENOIDS				
Lutein	5	881,922	4,765	*0.77 (0.68-0.87)*
Zeaxanthin	2	781,129	2,616	*0.79 (0.67-0.93)*
Beta Carotene	5	881,922	4,765	*0.98 (0.70-1.36)*
VITAMINS A, C, D, E, K				
Vitamin E	4	127,855	2,404	*0.67 (0.45-0.99)*
Vitamin C	5	128,222	2,559	*0.76 (0.61-0.94)*
Vitamin D	2	4,112	767	*0.79 (0.65-0.95)*
B VITAMINS				
Vitamin B1	2	4,597	1,909	*0.71 (0.55-0.92)*
POLYPHENOLS & FLAVONOIDS				
Tea	4	804,643	2,861	*0.71 (0.51-1.00)*
FATTY ACIDS & LIPIDS				
Polyunsaturated Fat PUFA	3	861,316	2,507	*0.94 (0.52-1.71)*
OTHER				
Alcohol	20	3,783,725	11,677	*0.72 (0.65-0.79)*
Total Dietary Fiber Intake	2	4,410	1,621	*0.85 (0.64-1.12)*



RR < 1 = negatively associated with renal cell carcinoma risk. | ◆ pooled RR (size reflects number of estimates) | ◇ red outline = statistically significant (95% CI excludes 1.0) | ◆ faded = not statistically significant | dashed line below CI = 95% prediction interval (where available) | N and case totals sum estimates and may overlap across strata | arrowhead = interval continues beyond the axis

Include subexposures: Yes